

GAYATHIRI SRIDHAR

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Computer Science undergraduate with hands-on experience building production-grade Retrieval-Augmented Generation (RAG) applications, AI-powered backend systems, and machine learning solutions. Passionate about Generative AI, backend engineering, and applied machine learning, with proven leadership across university innovation and research initiatives.

EDUCATION

B.E. Computer Science Engineering - Kingston Engineering College [CGPA: 9.16]

Class XII, CBSE - Spring Days Senior Secondary School [79%]

Class X, ICSE - BMD Jain School [88%]

EXPERIENCE

Generative AI Intern — Titan Company Limited [June 2026]

- Developed a production-grade Retrieval-Augmented Generation (RAG) chatbot for Titan's Taneira division, enabling semantic search across enterprise documentation and reducing manual knowledge lookup.
- Built a standalone ingestion pipeline supporting PDF parsing, chunking, embedding generation, vector indexing, and incremental knowledge-base updates without redeploying the chatbot.
- Tech Stack-Python, FastAPI, LangChain, Sarvam AI, PyMuPdf, PostgreSQL (pgvector), Docker, Uvicorn.

PROJECTS

Persona - Mental Health Support Chatbot for Students

- Built an AI companion bot giving students a private, accessible first point of contact for mental health support, designed to test stress levels using DASS21 test and route students to counselors when needed.
- Designed the chatbot workflow, prompt engineering pipeline, user experience for student interactions along with the research work
- Tested with 100+ undergraduate students across 1st–4th year, using feedback to improve usability and response quality.

Evoshield - AI-Powered Adaptive Firewall (in progress)

- Designing an adaptive firewall that uses a deceptive sandbox environment to trap and analyze mutating, AI-generated malware in real time.
- Responsible for the ML detection component and sandbox trap logic

MEMORA - Wearable Safety Monitor for Dementia Patients

- Built a smartwatch-based support system tracking fall events and real-time location for dementia patients and caregivers.
- Developed the backend APIs and integrated ESP32 hardware for real-time monitoring and alert generation.

Snipcraft - Browser-Based Video Toolkit

- Contributed to a browser toolkit for recording, editing, compressing, converting, and exporting video without switching apps.
- Built the recording and conversion module using an FFmpeg-based pipeline compiled to WebAssembly (WASM).

TECHNICAL SKILLS

- **Languages:** Python, Java, HTML, CSS
- **AI/ML & LLMs:** LangChain, LangGraph, Scikit-learn, PyTorch, TensorFlow, CatBoost, LightGBM, XGBoost, Hugging Face, RAG, Embeddings, Vector Search, AI Agents, MCP
- **Data Science:** Pandas, NumPy, NLTK, Matplotlib, Data Preprocessing, Feature Engineering
- **Backend & Databases:** FastAPI, REST APIs, PostgreSQL, pgvector, Firebase
- **DevOps & Tools:** Docker, Git, GitHub, Node.js, Jupyter, Google Colab, n8n
- **Research:** Literature Review, Scientific Writing, Technical Writing, Information Literacy
- **Design:** Figma, Canva

LEADERSHIP & ACHIEVEMENTS

- President, Innovation Institution Council (2025–Present) [organized various events under IIC for UG students]
- Editor, Research and Development Cell [Designed and composed contents of College magazine]
- Student Coordinator, SPEKECTRA (national-level technical symposium) [Kingston Engineering College]
- 1st Place, AATRAL'25 Presentation, VIT Math Fest
- 1st Place, StartBiz Business Plan Competition
- Winner, Project Expo '26 [Kingston Engineering College]
- Top 10, Hacksprint (36-hour hackathon) [4000+ participants]
- Participant, 24-hour Hackathon.